

Lecture - 9

- Point Estimator

(unknown populaⁿ param. θ could be μ, σ^2, p .
 $\rightarrow$ gives us 1 no. $\hat{\theta}$, $\hat{\mu} = \bar{X}$, etc.
 when is the point estimator good

$\rightarrow$ unbiased, SE, MSE, consistency, bias-var. trade off

- Unbiasedness $\rightarrow$ An estimator $\hat{\theta}$ is unbiased θ if $E[\hat{\theta}] = \theta$. The estimator is correct on avg.

eg. $\bar{X}$ an unbiased estimator of μ .

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad E(X_i) = \mu, \text{ then } E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

$$E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \times n \mu = \mu, \quad \text{yes}$$

- populaⁿ variance estimaⁿ

$$\frac{(n-1) S^2}{\sigma^2} \sim \chi_{n-1}^2 \quad \text{where } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

for chi square $Y \sim \chi_v^2 \Rightarrow E[Y] = v$

$$E\left[\frac{(n-1) S^2}{\sigma^2}\right] = n-1 \Rightarrow \frac{(n-1)}{\sigma^2} E[S^2] = (n-1)$$

$$E[S^2] = \sigma^2$$

what if we divide by n instead of $(n-1)$.

$$E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2, \quad \text{not equal to } \sigma^2$$

$\therefore \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is biased for σ^2 .

- Bias($\hat{\theta}$) = $E[\hat{\theta}] - \theta$ for unbiased, bias($\hat{\theta}$) = 0
 for above eq. Bias($\hat{\sigma}^2$) = $\frac{(n-1)\sigma^2}{n} - \sigma^2 = \frac{n\sigma^2 - \sigma^2 - n\sigma^2}{n}$

$$\text{Bias}(\hat{\sigma}^2) = \frac{-\sigma^2}{n}, \quad \text{we have a neg. bias here.}$$

$$- \boxed{SE(\hat{\theta}) = SD(\hat{\theta}) = \sqrt{Var(\hat{\theta})}}$$

measures variability
of an estimator
across repeated samples

measures variability
of indi. obs.

$$Var(\bar{X}) = \frac{\sigma^2}{n}, \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\text{if } \sigma \text{ is unknown } SE(\bar{X}) = \frac{S}{\sqrt{n}}$$

increase the n , makes our estimate more precise.

- We have $\hat{\theta}_1$ and $\hat{\theta}_2$ both unbiased,
 $E[\hat{\theta}_1] = E[\hat{\theta}_2] = \theta$, but if $Var(\hat{\theta}_1) < Var(\hat{\theta}_2)$
we choose $\hat{\theta}_1$ because its sampling
distribution is more concentrated around θ .
⇒ An unbiased estimator with smaller
variance produces estimates which lie
closer to the truth.

$$- \text{MSE} \Rightarrow \boxed{MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]}$$

$$\text{can be rewritten as } \boxed{MSE(\hat{\theta}) = Var(\hat{\theta}) + [Bias(\hat{\theta})]^2}$$

$$\text{if } Bias(\hat{\theta}) = 0, \text{ then } MSE(\hat{\theta}) = Var(\hat{\theta})$$

- Consistency → An estimator T_n of θ is
consistent if $T_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$.
As sample size become very large, estimator
becomes close to the true parameter.

$$E \Rightarrow \boxed{P(|T_n - \theta| > \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty}$$

$$P(|T_n - \theta| < \epsilon) \rightarrow 1$$

- Low bias & low var $\rightarrow$ tightly grouped around the target. Ideal case
- Low bias & high var $\rightarrow$ centered around the target but scattered. Estimator is correct on avg. but unstable.
- High bias & low var $\rightarrow$ grouped tightly but at wrong locaⁿ.
- High bias & high var. $\rightarrow$ worst case.
- some bias is acceptable, since final criterion is MSE
- Lasso $\rightarrow$ biased estimator but small var. (regularizaⁿ)
- Least sq. (ideal) $\rightarrow$ unbiased esti. + min. var.

- Interval Estimator $\rightarrow \hat{\theta}_L < \theta < \hat{\theta}_U$
 - $\rightarrow$ it tells approx. where the estimator lies
 - $\rightarrow$ also, how uncertain our estimate is.

$$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha \quad \rightarrow \text{confidence coefficient} \quad 0 < \alpha < 1$$

Then $[\hat{\theta}_L, \hat{\theta}_U]$ is c/d a $\boxed{100(1-\alpha)\%}$ CI.
confidence level

like 95% CI $\Rightarrow 1 - \alpha = 0.95 \Rightarrow \alpha = 0.05$

99% CI $\Rightarrow 1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$

$\alpha \downarrow$, CI $\uparrow$

- A 99% CI gives much greater confidence than 95% CI. To achieve greater confidence, we need a larger interval.

Higher confidence $\rightarrow$ wider interval
 ideal, high confidence + short interval
 confidence $\uparrow \Rightarrow$ interval width $\uparrow$
 but narrower interval $\Rightarrow$ more precision

Ex. $I_1 = (5'2, 5'4)$ $I_2 = (5'0, 5'6)$, $1 - \alpha = 0.95$
 which to prefer?

$L_1 = 2$, $L_2 = 6$
 $L_1 < L_2$

as same $\alpha (0.05)$, lesser wide (L_1) is
 more precise.